

Shi Chen

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EDUCATION

- ETH Zurich** Sep. 2025 – Present
Ph.D. in Mechanical and Process Engineering | Advisor: Prof. Stefan Leutenegger Zurich, Switzerland
- ETH Zurich** Sep. 2022 – Jun. 2025
M.Sc. in Electrical Engineering and Information Technology | GPA: 5.71/6.0 Zurich, Switzerland
- **Courses:** 3D Vision, Image Analysis and Computer Vision, Mixed Reality, Vision Algorithms for Mobile Robotics, Robot Learning, Computer Vision and Artificial Intelligence for Autonomous Cars, etc.
- Kyoto University** Apr. 2018 – Mar. 2022
B.Eng. in Electrical and Electronic Engineering | GPA: 3.94/4.3 Kyoto, Japan
- **Courses:** Machine Learning, Pattern Recognition, Computer Software, etc.

RESEARCH EXPERIENCE

- ProDyG: Progressive Dynamic Scene Reconstruction from Monocular Videos** Nov. 2024 – May 2025
NeurIPS 2025 & ETH Medal 2026 Advised by Dr. Erik Sandström and Prof. Martin Oswald ETH Zurich
- Devised online dynamic scene reconstruction pipeline from unposed monocular videos using 3D Gaussian Splatting;
 - Demonstrated robust tracking and novel view synthesis performance on par with state-of-the-art offline methods.
- MonoDy-GS: Online Monocular Dynamic Gaussian Splatting** Apr. 2024 – Aug. 2024
Advised by Prof. Martin Oswald, Dr. Sandro Lombardi and Prof. Marc Pollefeys ETH Zurich
- Developed online pipeline capable of modeling dynamic scenes from RGB-D sequence using 3D Gaussians;
 - Incorporated various priors based on real-world scene dynamics to address lack of multi-view coverage.
- EvenNICER-SLAM: Event-based Neural Implicit Encoding SLAM** Oct. 2022 – Jan. 2023
Advised by Dr. Danda Pani Paudel and Prof. Luc Van Gool ETH Zurich
- Enhanced robustness of NICE-SLAM against fast camera motion by integration of event data input;
 - EvenNICER-SLAM significantly outperforms NICE-SLAM in tracking with reduced RGB-D input frequency.
- Road Scene Pedestrian Relocation for Data Augmentation** May 2021 – Feb. 2022
Advised by Prof. Shohei Nobuhara and Prof. Ko Nishino Kyoto University
- Devised data augmentation method that automatically cuts out large-scale pedestrians in foreground of road-scene videos and relocates them at farther positions with correct scale and occlusion;
 - Proposed method improves Mask R-CNN in both detection and instance segmentation of far-away pedestrians.

SELECTED PROJECTS

- NeRaser: NeRF-based 3D Object Eraser | Mixed Reality** Oct. 2023 – Jan. 2024
- Developed object removal pipeline in NeRF-based scene representation trained on casual monocular videos;
 - NeRaser ensures visual consistency of occluded area through 3D-aware inpainting strategy.
- Monocular Visual Odometry | Vision Algorithms for Mobile Robotics** Nov. 2023 – Jan. 2024
- Implemented monocular feature-matching-based visual odometry pipeline in Matlab.
- Hierarchical Dense Neural Point Cloud-based SLAM | 3D Vision** Mar. 2023 – Jun. 2023
- Enhanced robustness of Point-SLAM through coarse-to-fine optimization using multiple levels of neural points;
 - Achieved significant improvement in tracking on real-world dataset ScanNet.
- Generalizable 3D Reconstruction Using Correspondence Matching | Robot Learning** Mar. 2023 – Aug. 2023
- Proposed generalizable SDF-based sparse-view 3D reconstruction method utilizing 2D feature matching.

COMPETITIONS & AWARDS

- ETH Medal 2026** May 2026
- Ichikawa International Scholarship** Oct. 2020 – Sep. 2021
- Champion, Summer Camp Robot Competition 2019 | Kyoto University** Sep. 2019
- Champion, Summer Camp Robot Competition 2018 | Kyoto University** Sep. 2018

SKILLS

Programming: Python, Matlab, PyTorch, Git, Docker, Singularity, Arduino

Languages: Chinese (native), Japanese (native), English (Fluent)